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▶ **To cite this version:**

Elodie Bouzekri, Torkil Clemmensen, Jose Abdelnour Nocera, Antony William Joseph, Arminda Guerra Lopes. Introduction to the Transparent Autonomy and Human Work Interaction Design Workshop. INTERACT 2025, Sep 2025, Belo Horizonte, Brazil. pp.91-98, ⟨10.1007/978-3-032-22937-3_8⟩. ⟨hal-05744886⟩

HAL Id: hal-05744886

https://hal.science/hal-05744886v1

Submitted on 10 Sep 2026

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Introduction to the Transparent Autonomy and Human Work Interaction Design Workshop

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Abstract. This introduction to the Transparent Autonomy and Human Work Interaction Design workshop presents the workshop's topic, its motivations, and the papers published on the proposed research theme. This workshop was collocated with the INTERACT2025 conference in Belo Horizonte, Brazil. The workshop proposed investigating the design of autonomous systems' transparency with the aim of enabling effective and pleasant collaboration with these systems at work. From the initial workshop papers, this volume presents extended and novel propositions to contribute to human work interaction design of autonomy transparency.

Keywords: Human Work Interaction Design · Transparency · Teaming · Collaboration.

1 Introduction

The workshop Transparent Autonomy and Human Work Interaction Design (TA-HWID) was organized by the Human Work Interaction Design (HWID) working group 6 of Technical Committee 13 of the International Federation for Information Processing (IFIP). The research interest of Human Work Interaction Design (HWID) is the design of usable and empowering systems that foster positive experience for workers by integrating interaction design methods and work analysis. For reference and a more detailed definition of HWID, one can refer to Clemmensen's book [7] or to the working group website[6].

The objective of the workshop on Transparent Autonomy and Human Work Interaction Design was to study transparency within teams that incorporate one or several autonomous systems through the lens of HWID [4]. More specifically, the workshop aimed to explore models, methods, and approaches for designing

[6] https://wg6.ifip-tc13.org/

autonomy transparency, considering the adaptation of workers' skills and tasks to work within these teams, and to collect case studies. The final objective of this workshop was to provide guidance on the human work interaction design of transparency within hybrid teams integrating autonomous systems.

To introduce the workshop, the organizers present the scope of the workshop's topic. This introduction was followed by a keynote given by Coppin [8] on the design of transparency to support collaboration in human-autonomy teams. Following the conclusion of this presentation and a discussion time, workshop papers were presented. This workshop proceedings volume presents a selection of extended versions of the papers and new contributions submitted by participants after the workshop, which draw on the workshop topics and discussions. Each papers included in this volume were single-blinded reviewed by two reviewers. These contributions enable the identification of gaps and research direction for team transparency in human work interaction design from design to implementation and evaluation of such systems.

2 Transparency

One factor that enables effective collaboration with autonomous systems is promoting an understanding of team members' roles, behavior, and accuracy. This concept is referred to as transparency [6, 13]. Transparency within a hybrid team, where both humans and autonomous systems work together, is concerned with making team members aware of each other's intents and sharing situation awareness for effective collaboration [3]. When referring to transparency, two main models are cited: Situation Awareness-based Agent Transparency (SAT) [5, 6] and the Lyons' model of transparency [13].

2.1 Situation Awareness-based Agent Transparency

SAT [5] is grounded in the Beliefs, Desires, and Intentions (BDI) agent framework [18], the Lee and See's "Purpose, Process, Performance" (3Ps) concept for trust in autonomous systems [12], as well as Endsley's theory of situation awareness (SA) [9]. Endsley defined three cumulative levels of SA. SA level 1 enables the human operator to perceive the environment [9]. SA level 2 allows the human operator to understand the situation [9]. At SA level 3, the human operator is able to project outcomes and future state of the situation [9]. SAT [5] defines three independent levels of transparency, where the information conveyed by the system to the human differs according to the three levels of SA. At SAT level 1, the system shares its current status, goals, actions, and plans [5], enabling the human operator to perceive the current state of work of the agent (i.e., what the system is currently doing and its plans). At SAT level 2, the system shares its reasoning process and the considered constraints, enabling the human operator to understand the situation [5]. At SAT level 3, the system provides a projection of its future state, along with associated uncertainty and anticipated consequences [5].

Later, Chen et al. [6] added bi-directional (i.e., from the system to the human and vice versa) communication of the team's decision-making processes and their shared tasks and interactions to their model.

2.2 Lyons' Model of Transparency

Lyons [13] defines two dimensions of transparency: 'robot-of-human' and 'robot-to-human' transparencies. Robot-to-human transparency refers to the information that an autonomous system needs to share with human operators [13]. Robot-to-human transparency encompasses the autonomous system's state, its understanding of the situation and the environment (i.e., intentional model, task model, analytical model, and environmental model). Robot-of-human transparency refers to the autonomous system conveying information related to its understanding and awareness of the human partner [13]. Robot-of-human transparency encompasses the communication of the system's understanding of the allocation of roles, as well as of the cognitive, emotional, and physical states of the human partner (i.e., the teamwork model and the human state model).

2.3 Transparency Within Human Teams

Transparency is also crucial between human team members for effective collaboration, especially in remote work settings where team members are not visible to each other, and communication is mediated by computers or other autonomous tools. Transparency can be conceptualized as a *disclosure of information* or as a *social process* [2]. In the first, transparency is a matter of quantity, quality, and relevance of information exchanges [2] (e.g., a worker reports to their superior that they are doing a particular task). In the second, the communication and mediation are emphasized [2] (e.g., how a conflict is resolved within a team). Considering the second, transparency refers to workers exchanging information about their thinking processes, tasks, issues, and the current state of work objectives to enable shared awareness and cooperation in remote settings [19]. Enabling such exchanges required the worker to be visible to others, understand shared information, and work ownership. They can also experience negative effects, such as surveillance or interruptions during their work [19].

3 Contributed Papers

Papers included in this volume investigate different aspects of transparency through three main tracks: design processes and methods, human-autonomy transparency, and human-human transparency. Three application domains are represented: the medical, maritime, and educational domains.

3.1 Design Processes for Human-Autonomy Teaming Transparency

Aladesuru et al. [1] highlight the benefits of user-centered design (UCD) of autonomous systems. Their study considered UCD of autonomous systems for

emergency care. They employed a combination of design thinking to explore the problem space and surveys. Their aim was to study users' acceptance of autonomous systems technology for emergency care in order to address the growing influx of patients. Their results showed the key importance of addressing transparency, trust, and user experience issues during the design process of medical autonomous systems. The authors found that including end-users in the design process revealed important contextual and work-related information that may be overlooked in a technology-centered process, such as wait-time transparency (i.e., waiting time before being seen by a doctor).

Advocating more towards co-design than UCD, Guerra [10] work aimed at drawing guidelines for the design of human-artificial intelligence (AI) collaboration in university education settings. To this end, this paper presents the results of interviews and focus group studies that investigate students' perceptions of AI tools and their impact on their study practices. A thematic analysis revealed that AI is perceived as a useful and efficient partner, despite highlighting accuracy and transparency issues that lead to distrust and a loss of ownership of the produced work. The analysis highlighted the importance of a well-designed transparency. The students reported "opaque decision-making", "misinformation", or "loss of authorship." Guerra [10] derived guidelines about transparency for HWID of AI for students, including: sharing cues on AI operations (e.g., rationale, uncertainty, sources of generated suggestions) and on AI and human roles, designing meaningful users' roles and fostering engagement (e.g., enabling interaction logging or previews of suggestions), or promoting culture-aware co-design with target users.

Directly involving end-users and other stakeholders in the whole design process, Joseph and Skyttä [11] present an adaptation of the Google Design Sprint to design a human-autonomous systems interface for managing fallback states (i.e., the autonomous vessel state when the current scenario is not included in its predefined plans) in the maritime domain. Instead of a 5-day design sprint, they experimented with a series of workshops over a 3-month period. Additionally, they investigated the inclusion of a facilitator role in Design Sprints. This study highlighted the facilitator's role benefits, which include their utility to facilitate participation in the group, providing structure to the Sprint process, and introducing new approaches to solve the design problem. Adding such a role was beneficial in developing emerging design ideas to support the transparency of autonomous systems.

3.2 Human-Autonomy Transparency

Still within the maritime context, Petermann et al. [17] present an evaluation of the Maritime Head-Up Display, which aimed to improve ferry operators' situation awareness in field settings. They implemented different transparency cues, such as bearing (i.e., direction from the vessel to an object of interest) indications through an LED strip, and light patterns (e.g., color, number of LEDs, blinking) to communicate information on the nature of the object or its status. Their results showed that coding bearing cues with an LED strip can enhance

situation awareness. However, encoding specific scenarios with light was difficult to understand (e.g., the target passes the vessel from behind), impairing level 2 of situation awareness-based agent transparency in such cases.

In the context of shore-based Remote Operation Centres, Petermann et al. [16] present a secondary analysis on a previous semi-structured interview study with navigators and gamers, which investigated their experiences with an interface for monitoring autonomous vessels. This interface displayed autonomous systems transparency notification to inform the user of the vessel's tasks, status, and requests to takeover control. The results of the thematic analysis were used to develop design guidelines aimed at improving transparency and situation awareness. Among the propositions, the authors proposed to wider camera angle for a more comprehensive view, avoiding blind spots, integrating the map and camera views, and using standardized symbology to code vessels' information.

3.3 Human Team Transparency

Transparency between human team members was also of interest to the workshop participants. Mughal et al. [15] propose exploring organizational transparency as "the visibility of actual work practices to both practitioners and system designers" in the medical context. Their paper presented an analysis of intensive care units' shift handover using the Functional Resonance Analysis Method (FRAM) and compared the prescribed protocol (Work-As-Imagined) to the reality of operations (Work-As-Done). This analysis helped identify key activities and information to convey within a team, promoting transparency and efficient coordination during handovers. Examples of information and resources exchanged in Work-As-Done, as revealed by FRAM, which are not described in protocols, include information shared with the patient's family and paper synthesis documents.

4 Discussion and Conclusion

The six workshop papers examine various aspects of transparency in work teams. Workshop participants proposed ways to ensure a required level of transparency, such as including target users as designers, considering cultural aspects, integrating a clear and unambiguous definition of actors' roles, task allocation, and anticipating how a close and repeated relationship would influence humans' skills and responsibility in work production. In addition, the participants highlighted that the design of coordination tasks is particularly critical during collaboration when a takeover or reallocation of control occurs. They proposed that the design process can benefit from systemic analysis methods (e.g., FRAM) and integrating a facilitator role in the design process.

Despite covering the entire spectrum of the design process, from methodological aspects to the design and evaluation of actual systems, some key dimensions of transparency are absent from the workshop contributions.

Projected future state: First, the communication of the future tasks, predicted outcomes, and associated constraints and uncertainty seems to be missing in the presented systems. However, the design recommendations of Guerra [10] include such a level of transparency corresponding to level 3 of situation awareness-based agent transparency.

Bi-directional communication: Another missing aspect is the bi-directional communication and the autonomous system awareness of the user's state. Little guidance is offered by the different contributions for retrieving the user's physical or cognitive states or to guide the frequency and the nature of the user's inputs required to maintain a common awareness of the situation. A human-centred design approach can help reduce new types of human errors in developing situational awareness.

Interaction time and frequency: It was discussed during the workshop that too much transparency may, on the contrary, decrease transparency. Transparency, which involves conveying information to team members, carries a high risk of overloading workers at a particular point in time. To avoid such pitfalls, Miller [14] proposed adding a time dimension to transparency.

Ethical considerations: Finally, an important dimension insufficiently addressed in the workshop contributions concerns the ethical implications of transparency, particularly in relation to human values and the disclosure of how autonomous systems negotiate or embed human and organizational values. As autonomous decision-making becomes more deeply integrated into work practices, understanding how systems represent, trade off, or potentially distort stakeholder values becomes crucial for maintaining trust and accountability. Transparency regarding these value-laden processes (and the limits of such transparency) remains an open challenge. Future research should therefore incorporate explicit ethical frameworks to support the responsible design of autonomy and to ensure that transparency does not only serve functional collaboration, but also fosters fairness, inclusiveness, and meaningful human oversight.

We, the organizers of the Transparent Autonomy and Human Work Interaction Design workshop, hope that the work presented in this workshop, along with the six papers included in the proceedings, will encourage future research on designing autonomous systems for the workplace while promoting transparency for effective and satisfying collaboration.

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